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Executive Summary

Ph.D. AI Executive & Principal Scientist with over 20 years of experience bridging the gap between cutting-edge research (NLP, Computer Vision, LLMs) and commercial ROI. Proven track record of architecting scalable, full-stack AI solutions for government (DARPA, IARPA, SOCOM) and enterprise (Grubhub, GE, Oculus360). Expert in securing multi-million-dollar R&D funding, managing global distributed teams, and deploying high-impact multi-modal Knowledge Graphs and state-of-the-art Deep Learning systems. Passionate about building research-driven cultures that consistently deliver production-grade software and tangible enterprise value.

Core Competencies

Strategic Leadership: R&D Strategy & Roadmap, Remote/Distributed Team Management, Grant Writing & Proposal Leadership (SBIR/DARPA/IARPA), Agile Data Science Methodologies, Cross-Functional Mentorship, Talent Acquisition.
AI & Natural Language Processing: Generative AI, Large Language Models (LLMs), RAG, LLM Fine-Tuning (Hugging Face, llama.cpp), Agentic Coding, Conversational AI (Chatbots), Transformers (BERT/GPT).
Core NLP & Computer Vision: Information Extraction (IE), Named Entity Recognition (NER), Semantic Parsing, Coreference Resolution, Sentiment Analysis, Bias Detection, Object Detection, Object Segmentation.
Knowledge Engineering: Knowledge Graphs (JanusGraph, TinkerPop), AI-Assisted Ontology Development, Multi-Modal Data Fusion, Ontology Learning.
Engineering, Data & MLOps:

- *Languages:* Python (Expert), Java (Expert), Scala, TypeScript, C++.
- *Big Data & Infrastructure:* Apache Spark, Hadoop, Kafka, ActiveMQ.
- *Web & Databases:* Next.js, Spring, PostgreSQL, Redis, DuckDB.
- *Frameworks & Cloud:* PyTorch, TensorFlow, Keras, Pandas, AWS, Hybrid-Cloud Architectures, Edge Computing.

Professional Experience

Dallas College | Richardson, TX
Faculty (Computer Science) | *2024 – Present*

As a Professor, I focus on shaping the technical foundation of the next generation of software engineers. I treat the classroom as a laboratory for refining technical communication, translating complex algorithmic concepts for diverse audiences, and identifying high-potential talent.

Key Contributions:

- **AI Leadership & Project Management:** Founded and serve as the Faculty Advisor for the campus AI Club, successfully scaling the organization to 70+ active members. Directed an

intensive AI summer project, instructing students in modern industry workflows including Agile methodologies, version control (Git/GitHub), Agentic coding, and end-to-end chatbot development.

- **Pipeline Development & Mentorship:** Mentor undergraduate students through rigorous technical training in C++, Java, and Python. Lead the AI effort of the UR@DC mentorship program, guiding students on the practical applications, limitations, and ethical considerations of Artificial Intelligence.
- **Curriculum Modernization:** Actively redesign course materials to bridge the gap between academic theory and enterprise reality by embedding AI-assisted coding and modern software engineering best practices into the core syllabus.
- **Technical Communication:** Refine instructional methods for explaining complex architectural and data structure concepts to non-experts, a skill directly translatable to executive stakeholder management and cross-functional team leadership in enterprise environments.

Core Courses Instructed:

- **ITSD 3330:** Data Structures and Algorithms
- **ITSD 3301:** Data Ethics
- **Programming Fundamentals I, II, & III:** C++, Java, Data Structures
- **ITSE 1370:** Introduction to Python

Core Skills and Technologies:

- **Engineering Practices:** Agile Methodology, Git/GitHub, Agentic Coding, AI-Assisted Software Development, Chatbot Development.
- **Languages & Concepts:** Python, Java, C++, Data Structures, Applied Algorithms, Data Ethics.
- **Leadership:** Technical Communication, Cross-Functional Mentorship, Curriculum Architecture.

Lang3s | Richardson, TX

Founder & Principal Architect | 2025 – Present

An Artificial Intelligence and Natural Language Processing (NLP) consultancy delivering state-of-the-art (SOTA) text analytics and bespoke AI product development. Lang3s bridges the gap between language, culture, and technology by transforming unstructured text into actionable intelligence. Alongside bespoke consultancy, Lang3s develops a proprietary NLP analytics platform that integrates advanced cultural AI models. This platform interprets high-volume data through multi-dimensional cultural and generational frameworks (e.g., Gen Z, Millennial, regional contexts), allowing organizations to uncover the nuanced sentiment and cultural drivers behind their data.

Core Capabilities:

- **Custom Model Building & SOTA NLP:** Designed and deployed advanced NLP pipelines specializing in custom extraction tasks, including claim detection, coreference resolution, entity recognition, and semantic parsing.
- **LLM Fine-Tuning & Optimization:** Led the fine-tuning and deployment of Large Language Models (LLMs) for domain-specific enterprise applications, utilizing PyTorch and the Hugging Faceecosystem for training, and optimizing local inference and quantization via llama.cpp.
- **Advanced Analytics & Cultural Intelligence:** Architected a proprietary AI analytics platform capable of extracting deep, human-centered insights and multidimensional cultural context from diverse, unstructured text sources.
- **Full-Stack AI Architecture:** Engineered scalable, end-to-end AI applications using a modern stack, bridging high-performance data processing with intuitive user interfaces.

Core Skills and Technologies:

- **AI/ML & NLP:** PyTorch, Hugging Face, Large Language Models (LLMs), llama.cpp, LLM Fine-Tuning, Custom NLP Models, SOTA NLP Processing.
- **Languages & Frameworks:** Python, Next.js.
- **Data & Infrastructure:** PostgreSQL, Redis, DuckDB.

Language Computer Corporation | Richardson, TX

Principal Scientist & Investigator
Full Time: 08/2018 – 07/2024 | **Part Time:** 08/2024 – Present

As a hands-on Principal Scientist and Investigator, I lead R&D for high-stakes AI programs (DARPA, SOCOM, Navy), driving the transition of theoretical research in NLP, Computer Vision, and Knowledge Graphs into mission-critical software. I bridge the gap between academic novelty and production-grade stability.

Key Contributions:

- **Massive-Scale Knowledge Graphs & Ontology Engineering (SOCOM):** Designed and deployed a 1B+ node multi-modal knowledge graph with a custom CRUD-enabled platform to integrate text, imagery, video, and speech. Pioneered AI-assisted ontology development to scale graph architecture. Solved critical edge-computing challenges by implementing secure subgraph dissemination protocols for disconnected field devices.
- **Computer Vision & Multi-Modal Analytics:** Developed and deployed custom AI models for object detection and object segmentation, enabling the automated, highly accurate extraction of intelligence from diverse image and video datasets.
- **Automated Intelligence & Dossier Generation (DARPA):** Built an autonomous system that generates comprehensive, Wikipedia-style dossiers from massive unstructured corpora. Engineered Deep Learning models for fine-grained Information Extraction (IE), including infobox prediction, source credibility assessment, and complex event sentiment analysis. Designed and integrated custom AI models specifically for bias detection across varying sources.
- **Grant Success:** Consistently secured multi-million-dollar R&D budgets (yielding 2x-5x ROI vs. cost) by architecting and authoring winning technical proposals for competitive SBIR, IARPA, and DARPA solicitations.
- **Mentorship & R&D Pipelines:** Directed end-to-end dataset development lifecycles and managed complex annotation pipelines for supervised and semi-supervised machine learning tasks.

Core Skills and Technologies:

- **AI/ML & NLP:** PyTorch, TensorFlow, Hugging Face, Large Language Models (LLMs), Natural Language Processing (NLP), Deep Learning, Bias Detection.
- **Computer Vision:** Object Detection, Object Segmentation, Multi-modal Data Fusion.
- **Knowledge Engineering:** Knowledge Graph Construction, AI-Assisted Ontology Development, Ontology Learning.
- **Languages & Frameworks:** Python, Java, Spring.
- **Architecture & Data:** Kafka, ActiveMQ, Distributed Graph Databases (JanusGraph/TinkerPop), Scalable ETL Pipelines, Distributed Systems.

Colorado Technical University | Online / Remote

Assistant Professor (Computer Science) | *2018 – 2022*

While leading commercial R&D initiatives, I served as faculty to stay at the bleeding edge of computer science theory and mentor emerging data scientists. This dual role allowed me to consistently bridge the gap between academic rigor and industry scalability.

Key Contributions:

- **Advanced Technical Instruction:** Delivered comprehensive graduate-level curriculum in Big Data Analytics and Parallel Computing, ensuring deep theoretical mastery of the distributed systems concepts utilized in high-scale commercial environments.
- **Talent Development & Engineering Standards:** Mentored diverse student cohorts in modern software construction, emphasizing and enforcing the "production-grade" Python and Java coding standards required by top-tier technology firms.
- **Curriculum Strategy & Modernization:** Adapted and redesigned course materials to reflect the rapid evolution of Data Science algorithms, moving beyond textbook theory to focus on real-world, applied enterprise data challenges.

Core Courses Instructed:

- **Graduate Level:** Fundamentals of Big Data Analytics, Algorithms for Data Science
- **Undergraduate Level:** Parallel Computing, Software Construction, Advanced Java

Core Skills and Technologies:

- **Data & Architecture:** Big Data Analytics, Distributed Systems, Parallel Computing.
- **Engineering & Languages:** Python, Java, Advanced Java, Software Construction, Production-Grade Coding Standards.
- **Leadership:** Graduate-Level Instruction, Curriculum Strategy, Technical Mentorship, Data Science Application.

Grubhub | New York, NY (Remote)

Data Scientist Team Lead | *2017 – 2018*

Directed the Natural Language Processing (NLP) and Taxonomy roadmap for the Search & Discovery platform. Focused on transforming massive volumes of unstructured, noisy menu data into structured, highly queryable assets to drive customer conversion and platform revenue.

Key Contributions:

- **Search Relevance & Revenue Generation:** Solved complex entity resolution challenges and spearheaded automated ontology development to normalize and categorize millions of dishes and restaurants. This directly improved search ranking precision, driving measurable, data-backed gains in Customer Retention and Gross Order Volume.
- **Pioneering AI & Conversational Search:** Championed early-stage AI product innovation by leading the R&D, architecture, and deployment of Conversational AI (Chatbots) ahead of industry trends.
- **Automated Information Extraction:** Engineered and deployed automated machine learning models for ingredient and nutritional extraction, significantly enriching user-facing metadata and improving dietary search filtering capabilities.
- **Global Engineering Leadership:** Successfully managed a highly distributed, cross-functional team of 8 ML researchers and software engineers across New York, India, and Ukraine. Established rigorous Agile workflows and a seamless remote collaboration culture to ensure continuous, high-quality product delivery.

Core Skills and Technologies:

- **AI/ML & NLP:** Natural Language Processing (NLP), Conversational AI (Chatbots), Automated Ontology Development, Entity Resolution, Information Extraction, Taxonomy.
- **Search & Infrastructure:** Search Ranking Algorithms, Apache Spark, AWS (Amazon Web Services), Big Data Processing, Distributed Data Pipelines.
- **Languages:** Python, Scala.

Oculus360 | Addison, TX

Vice President of Technology | *2014 – 2017*

Led the technical evolution of a marketing analytics startup from an early-stage concept into a highly scalable, revenue-generating commercial AI platform for Fortune 500 enterprise clients. Operating as a hybrid Chief Architect and Engineering Leader, I successfully bridged the gap between deep Natural Language Processing (NLP) research and enterprise-grade, high-availability product delivery.

Key Contributions:

- **Enterprise-Grade Commercial Scaling:** Architected and launched a scalable, B2B commercial platform that translated abstract social data signals into actionable marketing intelligence. Directly drove client revenue growth and enterprise adoption by enabling highly targeted, data-driven consumer campaigns.
- **Proprietary AI & Psycholinguistics Engine:** Designed and deployed the core NLP and machine learning algorithms. Advanced platform capabilities far beyond basic sentiment analysis by engineering models to extract complex "Usage Occasions," nuanced emotional states, and distinct personality traits from millions of unstructured social interactions.
- **Big Data Infrastructure & Pipelines:** Engineered a high-throughput, horizontally scalable data architecture using Apache Spark and Hadoop. Enabled the near real-time processing of massive, high-velocity unstructured social corpora to meet strict enterprise client Service Level Agreements (SLAs).
- **Engineering Leadership & Culture:** Built the data science and engineering organizations from the ground up. Established production-focused engineering standards, Agile methodologies, and rigorous technical hiring pipelines to foster a high-performance, research-driven environment.

Core Skills and Technologies:

- **AI/ML & NLP:** Natural Language Processing (NLP), Machine Learning, Psycholinguistics, Sentiment Analysis, Applied AI.
- **Data & Infrastructure:** Apache Spark, Hadoop, AWS (Amazon Web Services), Big Data Architecture, High-Throughput ETL Pipelines.
- **Languages & Engineering:** Java, Scala, Python, Agile Methodologies, Enterprise SaaS Architecture.

Language Computer Corporation | Richardson, TX

Senior Scientist | *2011 – 2014*

As a Senior Scientist and Principal Investigator, I pioneered early enterprise approaches to Computational Sociolinguistics. By bridging the gap between linguistic theory and machine learning, I engineered advanced NLP systems capable of decoding complex human intent, social roles, and cultural dynamics across multiple languages.

Key Contributions:

- **Cross-Lingual NLP & Sociolinguistics (IARPA SCIL):** Served as Principal Investigator (PI) for the IARPA Sociocultural Content in Language program. Directed a cross-functional team of 10 ML researchers, software engineers, and linguists to develop novel algorithms for analyzing discourse structure and social hierarchy in English, Chinese, and Arabic.
- **Cognitive Modeling & Semantic AI:** Operated as a key technical contributor to high-profile DARPA and IARPA initiatives focused on complex Metaphor Detection. Engineered multilingual AI models to identify, extract, and interpret figurative language across English, Russian, Farsi, and Spanish unstructured corpora.
- **Behavioral & Cultural Analytics:** Innovated novel machine learning methodologies for Social Action Recognition and conceptual space induction. Advanced system capabilities to move beyond literal translation, allowing models to accurately infer the underlying intent and cultural context driving communications.

- **Executive Stakeholder Communication:** Successfully defended R&D pipelines and presented highly technical AI findings directly to the Office of the Director of National Intelligence (ODNI) and at executive IARPA program reviews.
- **Academic Excellence & Thought Leadership:** Authored and co-authored 22 peer-reviewed publications on NLP, Machine Learning, and Sociolinguistics, cementing the organization's reputation for cutting-edge innovation within the global academic and defense communities.

Core Skills and Technologies:

- **AI/ML & NLP:** Natural Language Processing (NLP), Computational Sociolinguistics, Cross-Lingual NLP, Machine Learning, Latent Variable Models, Sentiment Analysis, Cognitive Modeling.
- **Domain Expertise:** Cultural Analytics, Social Action Recognition, Conceptual Space Induction, Metaphor Detection, Discourse Analysis.
- **Languages Modeled:** English, Mandarin Chinese, Arabic, Russian, Farsi, Spanish.
- **Leadership & R&D:** Principal Investigator (PI), Defense R&D (DARPA, IARPA, ODNI), Grant Execution, Multidisciplinary Team Management.

General Electric Global Research | Niskayuna, NY

Computer Scientist | *2008 – 2011*

At GE's centralized R&D hub, I pioneered the development of advanced social media analytics, text mining, and web intelligence platforms. I successfully bridged the gap between R&D and enterprise commercialization, transitioning complex research prototypes into high-scale, revenue-generating tools for major media conglomerates (NBC Universal) and the defense sector.

Key Contributions:

- **Defense & Intelligence (WISDOM Program):** Served as Lead Architect for the Web Information Spread Data Operations Module (WISDOM). Led foundational research in text mining and social network analysis.
    - *Commercial Impact:* Directed the system's architecture and successful commercialization, leading to its acquisition by Lockheed Martin. The platform (LM WISDOM®) evolved into industry-leading predictive analytics and Open-Source Intelligence (OSINT) tool used for government security risk assessments, monitoring global political instability, and mitigating enterprise insider threats.
- **Media Analytics & Predictive Modeling (NBC Universal):**

- *Virality Prediction:* Engineered early, innovative "pre-viral" detection algorithms to automatically identify online communities and social networks with the highest propagation potential for Universal Pictures video content.
- *Audience Engagement:* Built the automated analytics engine for NBC.com. Extracted and mapped key behavioral drivers, enabling optimized content strategies that drove highly measurable, significant increases in user retention and overall Page Views.
- *Partner Identification:* Designed and deployed ranking algorithms to autonomously identify and evaluate high-value communities for strategic commercial partnership opportunities.

Core Skills and Technologies:

- **AI/ML & Data Science:** Predictive Analytics, Text Mining, Web Mining, Social Network Analysis (SNA), Pre-Viral Detection Algorithms, Behavioral Drivers.
- **Domain Expertise:** Open Source Intelligence (OSINT), Threat Identification, Commercialization of R&D, Audience Engagement, Predictive Modeling.
- **Languages & Architecture:** Java, Scalable Architecture, Automated Analytics Engines, Big Data Infrastructure.

Education

Ph.D., Information Science and System Engineering
The University of Tokushima, Japan

M.S., Computer Science
The University of Central Florida

B.S., Computer Science
The University of Central Florida

Selected Intellectual Property & Patents

*Inventor on **8+ U.S. Patents** focused on NLP, Consumer Analytics, and Intent Mining.*

- **Occasion-Based Consumer Analytics:** Methods for extracting context of use from unstructured text.
- **System and Method for Intent Mining:** Automated identification of user intent in dynamic social systems.
- **Multi-Factor Perception Analysis:** Advanced sentiment and perception modeling for brand analysis.

Selected Publications

*Author of **75+ peer-reviewed publications**. Full citation list available at www.davidbracewell.com.*

1. **Bracewell, D. B.** (2015). "Long Nights, Rainy Days, and Misspent Youth: Automatically Extracting and Categorizing Occasions Associated with Consumer Products." *SocialNLP 2015* (**Best Paper Award**).
2. **Bracewell, D. B.** (2015). "A Four-Factor Model for Mining Consumer Insights in Social Data." *Proceedings of the 28th International FLAIRS Conference*.
3. Mohler, M., **Bracewell, D. B.**, et al. (2013). "Semantic Signatures for Example-Based Linguistic Metaphor Detection." *Proceedings of the First Workshop on Metaphor in NLP*.
4. **Bracewell, D. B.**, et al. (2012). "A Motif Approach for Identifying Pursuits of Power in Social Discourse." *IEEE Sixth International Conference on Semantic Computing*.